Puszka Pandory

Final exam substitute

30 points total

Program

Reverse engineer an old Polish game ”Puszka Pandory”, then re-write the game in Prolog. The goal is to play the game and analyse its world and possible interactions. Reading the literature about the game is recommended.
The game file is provided to you. Use an emulator to run it (recommended Speccy for Windows/Linux and Fuse for Linux).

To receive a passing grade (16/30 points), the following functionality is **required**:

1. Moving between locations - free motion around the world with few constraints
2. Looking around - exploring the current location
3. Picking up / dropping items, listing carried items in the inventory
4. Use of items to interact with the world in a way that causes a permanent change to the world. For example, using a shovel should create a hole in the ground that will persist even when the player leaves the location and can be found by coming back to the location.
5. At least 50% of biomes from the original game.

A full recreation of the original game is not required to obtain the maximum number of points, but the implementation should be **significantly** more detailed that what is described in the minimum requirements. Example extra functionality: More items, being able to die, graphics, storyline, full game map, etc.

Reverse-engineering can be done in **groups up to 4** students.
Programming must be done **individually** by each student.
Submitting plagiarised code is grounds for failing the project.

Report

Prepare a report consisting of two parts

1. Reverse engineering

Describe your play-testing experience. List all information that you have found about the game, link to source materials. Describe the world and possible interactions. You should reference this description in the next section.
Specify whether the work was done alone or as a group (list members of the group if applicable).

2. Recreation

Describe your implementation. Show examples of your implementation of the minimum requirements. Separately, show the examples of your implementation that goes beyond the minimum requirements. Reference how your implementation is recreating the original game by referencing the previous section.
Any changes to the functionality of the base game must be mentioned (e.g. if you combine MOVE and SWIM into a single command MOVE, this must be mentioned).

This part of the report should mention what Prolog version the game is written in as well as how to run it. If your game includes graphics, state the recommended terminal size (rows/cols).

Deliverables

The source code of the game must be delivered as a single `.pl` file.
Name the file `sXXXXXMIWFinal.pl` where `sXXXXX` is your student number.

The report must be delivered as a single `.pdf` file.
Name the file `sXXXXXMIWFinalReport.pdf` where `sXXXXX` is your student number.
Submissions missing the report will not be accepted.